

Software Requirements & Scope Agreement

Order Management System for Retail Clothing Store (Client: D'fine)

1. Document Details & Parties Involved

Project Title	Clothing Store Order Management System
Document Version	Version 1.0 (Requirements & Scope Baseline)
Technology Stack	• Backend: Java, Spring Boot (Spring Data JPA, Spring Security, RESTful APIs) • Frontend: React.js (with Vite), HTML, CSS , JavaScript • Database: PostgreSQL • Build & Tools: Apache Maven, Git
Date & Academic Term	Year 2, Semester 1, 2026
Client Organization	D'fine
Development Team	• IT25102762 - Yapa R. M. J. O. • IT25101865 - Fernando T. K. D. • IT25102773 - Nimradha G. S. • IT25102756 - Dilmith P. H. S.

2. Project Overview & Business Goals

The primary objective of this project is to engineer a robust, automated, and secure web-based Order Management System (OMS) tailored specifically for D'fine, a contemporary retail clothing store. The system modernizes retail store operations by eliminating error-prone manual sales logging, streamlining inventory tracking across garment variants, optimizing multi-tier order processing, and providing real-time transparency for both store management and customers.

- **Customer Shopping Experience:** Provide customers with an intuitive, fully responsive storefront to explore clothing apparel, filter items by size/color/category, configure shopping carts, apply promotional vouchers, checkout seamlessly, and track delivery progress in real time.
- **Inventory & Apparel Variant Control:** Empower store administrators to manage multi-attribute clothing variants (Size, Color, SKU), track dynamic stock levels, manage restock batches, and prevent overselling using atomic transaction boundaries.

- **Order & Fulfillment Automation:** Centralize order processing, payment verification (bank slip inspection) and streamline courier tracking from dispatch to delivery.
- **Customer Behavior & Trend Intelligence:** Equip store management with visual analytics to monitor customer registration growth, repeat customer retention trends, demographic purchasing preferences, and most demanded apparel categories and sizes.

3. System Functional Modules & Scope Allocation

In strict accordance with academic evaluation standards, the system is partitioned into 4 core functional modules of balanced complexity and equal scope. Each module encompasses full-stack responsibilities (front-end UI, backend services, controller APIs, and database persistence) with strict individual accountability:

Module & Domain	Key Functional Requirements & Scope	Responsible Member
1. User Management & Customer Behavior Trend Analytics	• User Authentication & Security: Registration, secure login/logout, session management, and password reset. • Role-Based Access Control: Admin Dashboard vs. Customer Profile. • User profile management: Shipping address, User details updates • Store Staff Administration: Admin interface to create store staff accounts, assign operational permissions. • Customer Behavior & Trend Analytics: Interactive visual dashboard visualizing product trends.	IT25102762 - Yapa R. M. J. O.
2. Products & Inventory Management	• Catalog & Category Management: browsing with category, size, color, and price filters. • Apparel Variant & SKU Tracking: Multi-attribute variant management (e.g.: Size: S, Color; Fit) with auto-generated unique SKU codes and individual variant stock levels. • Storefront Browsing & Multi-Facet Filters: Customer catalog filtering by category, size, color, and price range, with badges ('In Stock', 'Low Stock', 'Out of Stock'). • Restock Logging & Safety Threshold Alerts: Stock restock batch logging, inventory history records, and automated low-stock dashboard alert notifications when variant inventory drops below threshold.	IT25101865 - Fernando T. K. D.

	• Customer Product Reviews & Ratings: Customer product rating (1-5 stars) and review submission CRUD with administrative moderation.	
3. Order Handling & Cart Management	• Shopping Cart Management: Shopping cart management (add, update quantities, remove items, clear cart) with real-time validation against available variant stock. • Checkout Workflow & Atomic Order Placement: Checkout workflow, order summary computation, and order placement. • Order Lifecycle & Tracking: Customer order history review and status tracking (Pending, Processing, Completed). • Order History & Downloadable Invoices: Customer order history and automated downloadable PDF Order Invoices. • Admin / Staff Order Entry: Assisted Order Entry portal where staff can log orders received via social media, ensuring online stock is in sync with inventory.	IT25102773 - Nimradha G. S.
4. Payment & Delivery Management	• Bank Transfer Slip Verification Queue: Verification portal to inspect customer-uploaded payment receipts, approve/reject payments, and automatically update payment status. • Dynamic Shipping Rate Calculator: Delivery fee computation engine based on destination province/district and package weight parameters. • Courier Dispatch & Tracking Management: Courier partner assignment, automated tracking reference number generation, and delivery stage progress updates. (Via automated bot with partnered courier's website) • Fulfillment Slips & Automated Notifications: Automated email notifications sent to customers at each payment and delivery milestone.	IT25102756 - Dilmith P. H. S.

4. Non-Functional Requirements

- **Performance & Response Time:** Page rendering times and REST API endpoints must respond within 2.0 seconds under standard concurrency.

- **Security & Authorization:** User passwords securely hashed with BCrypt (strength factor 12). Administrative routes and APIs guarded by Spring Security RBAC. Defenses against CSRF, XSS, and SQL Injection via Hibernate/JPA parameter binding.
- **Data Integrity & Concurrency Control:** Inventory deductions, coupon redemptions, and order placements must execute within ACID-compliant database transactions (@Transactional) with optimistic locking to prevent overselling race conditions.
- **Usability & Responsive Design:** Web interfaces must be fully responsive across mobile, tablet, and desktop viewports, ensuring an accessible UI/UX for both customers and store administrators.
- **Maintainability & OOP Architectural Standards:** The application must strictly follow a 3-tier layered architecture (Controller, Service, Repository, Entity), DTO pattern, separation of concerns, and clean Object-Oriented Design principles.

5. System Assumptions, Dependencies & Client Responsibilities

To protect project feasibility and ensure high-quality implementation within academic deadlines, the following technical and operational boundaries are agreed upon:

- **Automated Credit/Debit Card Payment Gateways:** Third-party automated card payment gateway is excluded from this web site. the system will process transactions via Direct Bank Transfer with customer.
- **Supplier & Vendor Procurement Management:** Supply-chain procurement, vendor database management, and automated Purchase Orders to garment manufacturers are excluded from the site. Inventory restocks are logged directly by store managers.
- **Distributed Multi-Warehouse Logistics:** The system manages inventory for D'fine's primary centralized retail store facility. Multi-warehouse inter-facility inventory routing is excluded.
- **Multi-Currency & International Shipping:** Foreign currency conversions (USD /EUR) and cross-border customs fulfillment are out of scope. The platform is configured strictly for domestic operations denominated in local currency (LKR) with island-wide courier fulfillment.
- **Native Mobile Applications (iOS & Android):** Development of native smartphone application packages for app stores is excluded.
- **Domain & Cloud Hosting Procurement:** The client (D'fine) is solely responsible for purchasing, maintaining, and providing the custom domain name (dfine.lk) and the production cloud/web hosting infrastructure required for deployment.
- **Infrastructure Billing & Third-Party Subscriptions:** All recurring third-party operational expenses are funded directly by the client. The development team is responsible solely for the initial deployment, configuration, and source code handover.

6. Project Deliverables & Academic Milestones

Milestone / Evaluation Phase	Key Deliverables & Documentation	Target Timeline
Phase 1: Project Details & Scope Agreement	Client details, stakeholder roles, project scope document, and individual functional responsibility matrix.	Late August

Phase 2: Initial Presentation	Requirements gathering evidence, initial presentation slides, stakeholder meeting minutes, and architectural domain models.	Early September
Phase 3: Progress Evaluation	50% working implementation (functional UI views, functional REST APIs, and working database schema), finalized use cases, and testing roadmap.	Late September
Phase 4: Final Report & Product Viva	Comprehensive Turnitin group technical report (UML diagrams, test results, code walkthrough), deployed Spring Boot web application, and individual Viva demonstration.	October

7. Client Approval & Sign-Off

By signing below, the client representative and the development team formally agree to the functional requirements, project scope, technical specifications, and milestones outlined in this agreement document.

Client Representative:	Development Team Lead:
Organization: D'fine Apparel Store	Project: Clothing Store OMS
Name:	Name:
_____	_____
Designation:	Student ID:
_____	_____
Signature:	Signature:
_____	_____
Date:	Date:
_____	_____